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# Magnetostratigraphy of the Permo-Triassic boundary section at Shangsi (Guangyuan, Sichuan Province, China)

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Continuous marine sedimentation characterizes many Late Permian to Early Triassic sections on the Yangtze terrane in South China. The Permo-Triassic (P/Tr) boundary section at Shangsi (Sichuan Province) consists of limestones intercalated with clays and mudstones which belong to the Wuchiapingian and Changxingian (Upper Permian) and the Griesbachian and Dienerian (Lower Triassic) stages. The P/Tr boundary is formed by a clay horizon with an unusually high iridium concentration. The intensity of natural remanent magnetization is very low with a mean of 0.15 mA $m^{-1}$. About 40% of the samples contain secondary or unstable magnetization components only, whereas the remaining samples carry a characteristic remanent magnetization thought to reflect the polarity of the geomagnetic field during deposition with sufficient accuracy. Normal and reversed polarity of the characteristic magnetization constitute a pattern of at least six polarity zones, the P/Tr boundary being situated very close to the transition from a reversed to a normal polarity zone. The Shangsi polarity sequence represents part of the Illawarra interval of mixed polarity, the exact beginning of which has still to be determined.

## 1. Introduction

A large number of terranes has been accreted during a complicated and long lasting amalgamation process to form the geographical outline of present day China [1]. Geological observations [2] and palaeomagnetic measurements [3] show that the major tectonic elements of South China, the Yangtze block and the South China fold belt (Fig. 1) did not suture together before the end of the Triassic. The Yangtze platform was consolidated during the Late Proterozoic Sinian orogeny [5], but has been deformed by the continued interaction between the Pacific, Indian and Eurasian plates [1].

During the Late Palaeozoic and the Early Mesozoic the Yangtze platform was transgressed by a shallow sea. Continuously deposited, marine Permo-Triassic (P/Tr) carbonate sequences are found at many places [6]. Contact conformity between the sediments of the Changxingian (Uppermost Permian) and the Griesbachian stage (Lowermost Triassic) and fossiliferous strata with a rich and stage characteristic fauna are typical of many P/Tr boundary sections on the Yangtze terrane. A clay horizon, which is situated at the bottom of a bed with mixed Permian and Triassic fauna elements, represents the P/Tr boundary layer [6]. In this clay layer anomalous abundance variations have been observed for Th, Ta, Hf and Cr [7] and for other siderophile elements and iridium [8,9]. Moreover, observations have shown high concentrations of microspherules [8] and a carbon isotope shift [10] similar to the one which occurred at the Cretaceous-Tertiary boundary. The results between studies are often contradictory. Iridium depletion has even been reported for the boundary clay [11]. Nevertheless most authors have interpreted the geochemical anomalies as resulting from an extraterrestrial impact event as the cause of the dramatic fauna turnover from the Paleozoic to the Mesozoic era.

The Permian (and part of the Upper Carboniferous) was a time of long lasting, reversed polarity of the geomagnetic field. This Kiaman reversed interval [12,13] was followed by the Illawarra

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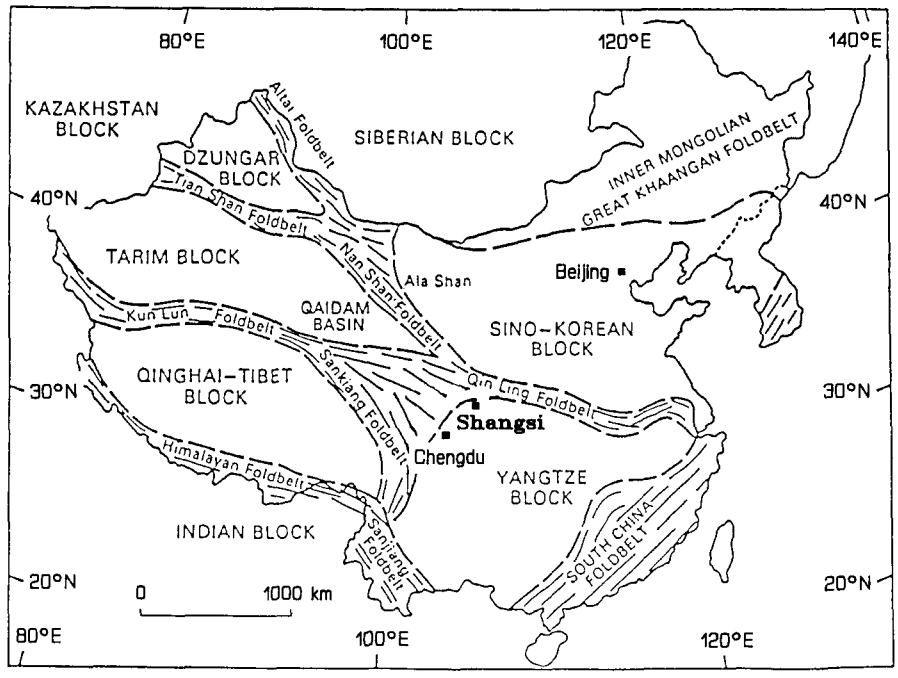



Fig. 1. Simplified tectonic map of China after McElhinny [4].

mixed polarity interval. The onset of these reversals is not well known [14] because the palaeomagnetic signal has been observed mainly in terrigenous red sediments [15,16] which are poorly dated and cannot easily be correlated worldwide.

The continuous marine carbonate sedimentation across the P/Tr boundary in many sections of the Yangtze platform provides a unique opportunity for obtaining a complete and palaeontologically well constrained magnetostratigraphic record of the palaeomagnetic field throughout this time. A monoclinal, moderately southeast dipping, P/Tr boundary type section is exposed at Shangsi near the city of Guangyuan (Sichuan province; 32° N, 105.5° E) at the northwestern corner of the Yangtze block. The nearly 200 m thick sedimentary sequence consists of well bedded, partly very pure micritic and sparitic, partly siliceous limestones intercalated with clays and mudstones. In the lower part they belong to the Dalong formation which represents the Changxingian stage and the upper portion of the Wuchiapingian stage. These stages together with underlying Maokou Formation encompass the entire Late Permian in China. The upper part of the section belongs to the Feixianguan formation of Lower Triassic age (Fig. 2) which incorporates the Griesbachian and at least part of the Dienerian stages. The P/Tr boundary here is also represented by a clay horizon with unusually high iridium concentration [9]. At the boundary all non-fusulinid foraminifera and radiolaria become extinct and the goniatite, ceratite and ammonoid families are reduced severely [17]. Some Permian fauna elements such as Pleuronodoceras or Pseudotirolites or small sized brachiopods survive and become accompanied with newcomers of Claraia. At the same time the sedimentary basin facies changes to a tidal flat facies. The basal layers of the early Triassic contain Hypophiceras, Claraia wangi and Claraia griesbachi, whereas Otoceras seems to be missing in southern China (A. Baud, oral communication). The absence of any obvious hiatus and the early appearance of a rich Mesozoic fauna promise a very complete record of geologic history. The Shangsi section seems well suited for a study of geomagnetic field polarity in the neighbourhood of the P/Tr transition.

## 2. Remanent magnetization

Nearly 200 samples were drilled at fairly regular intervals across 130 m of Wujiapingian to

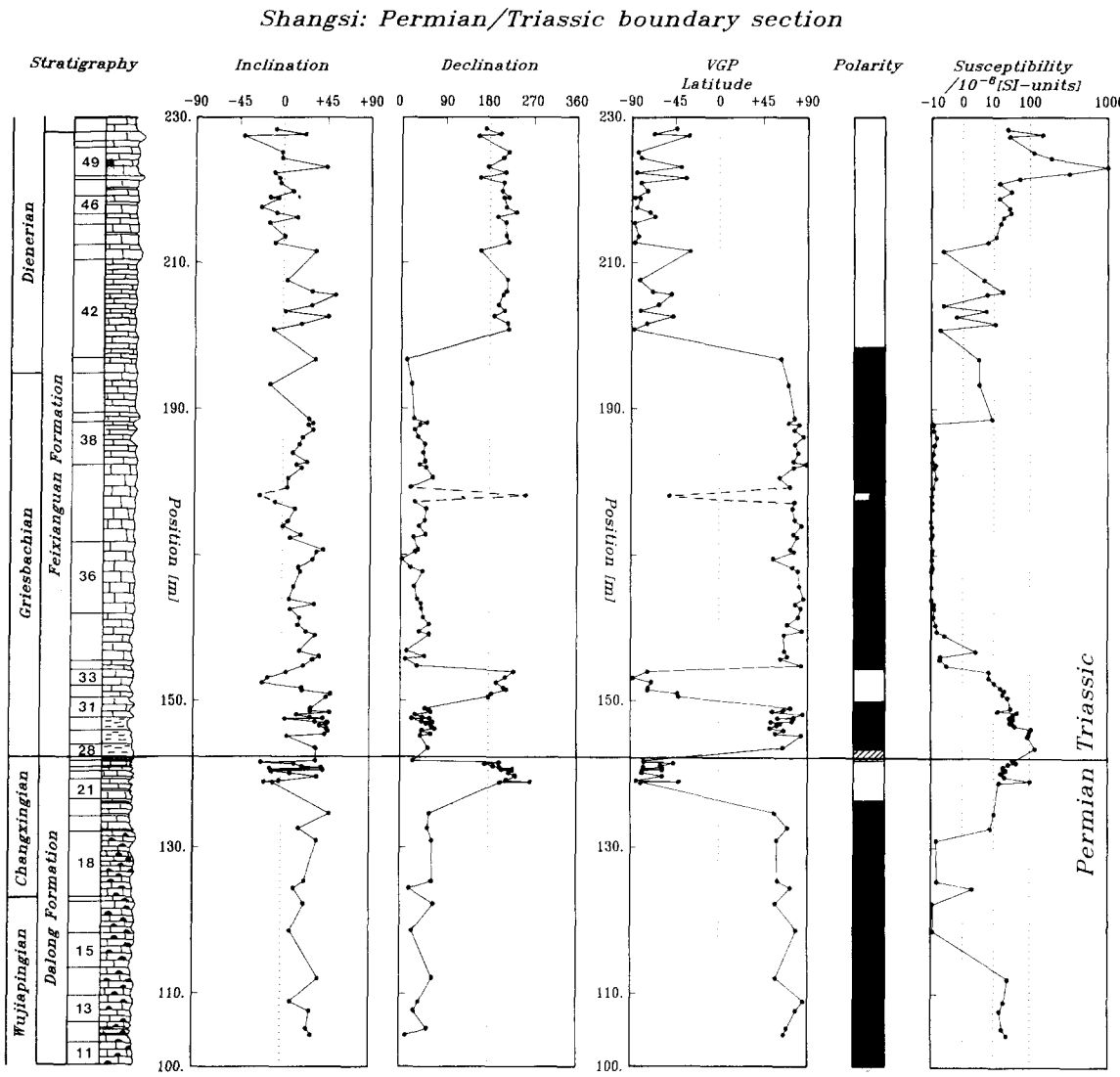



Fig. 2. Magnetostratigraphic section across the Permo-Triassic boundary at Shangsi (Sichuan Province, China). Stratigraphy column describes stratigraphic stages, regional formation names and lithology with thin and thick bedded limestones in the Triassic, siliceous limestones in the Permian and clays (black) and mudstones (dashed) around the P/Tr boundary and with the division numbers as assigned by the Chinese Academy of Geological Sciences and cited by Li et al. [17]. Inclinations and declinations of characteristic remanent magnetization are given as function of profile depth. The latitudes of the corresponding virtual geomagnetic poles (VGP) are represented as deviations from the mean pole which plots at an axial position. Normal polarity is black, reversed polarity white, uncertain is cross-hatched, single samples with opposite polarity are indented. The initial volume susceptibility is plotted proportionally to the cubic root of susceptibility.

Dienerian limestones (Fig. 2). In comparison with the red siltstones and limestones of the Lower Triassic Yelang Formation which was deposited a few hundred kilometres to the south [3], the Dalong and Feixianguan limestones at Shangsi are extremely weakly magnetic, with the mean intensity of natural remanent magnetization (NRM) averaging only 0.15 mA $m^{-1}$ (Fig. 3b). This is due to the very low concentration of ferromagnetic minerals. Even paramagnetic minerals are scarce, resulting in diamagnetic rock susceptibilities across major parts of the section (Fig. 2). In addition, the majority of NRM directions align close to the local dipole field direction (inclination: 51°), indi-

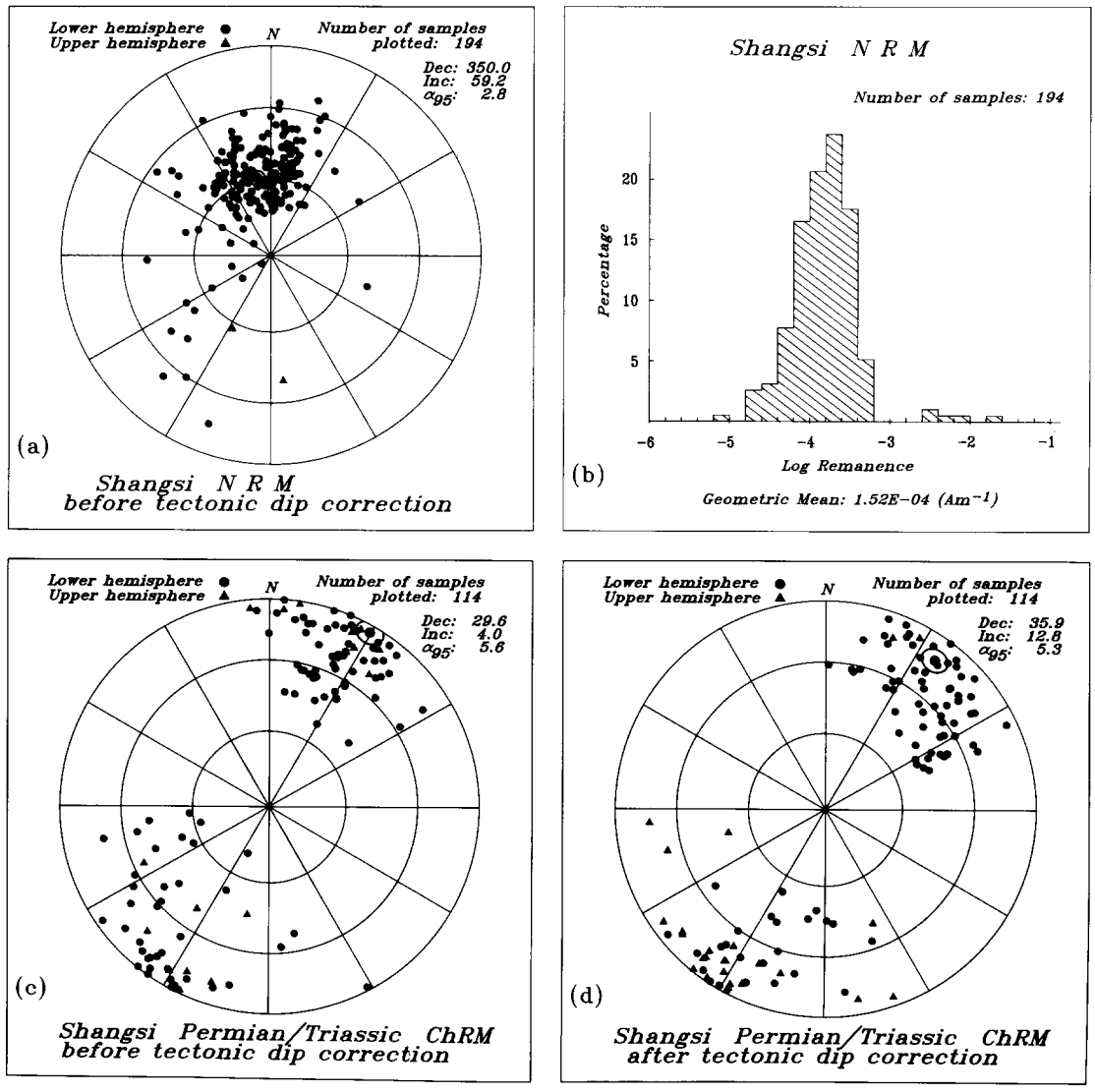



Fig. 3. Properties of natural remanent magnetization (NRM). (a, b) Directions and intensities of NRM before cleaning. (c, d) Thermally cleaned characteristic remanence directions before and after tilt correction. Figures of the overall mean directions including 95% circle of confidence [18] are given as well. The mean directions for normal- and reversed-polarity samples are not strictly antipodal due to overprint magnetizations in some reversed samples which could not be removed completely by thermal demagnetization.

cating that large NRM proportions are of secondary origin (Fig. 3a).

Thermal demagnetization of NRM and isothermal remanent magnetization (IRM) acquired in fields up to 1 T have been used to evaluate the nature of the magnetization components and their carrier minerals [19,20]. A characteristic remanent magnetization (ChRM), i.e. the magnetization component which trends to the origin of orthogonal vector demagnetization diagrams, could be determined—although sometimes with great difficulty—for about 60% of the samples (Fig. 3c, d). The remaining 40% had to be discarded because of instability, NRM component overlap, complete (viscous) remagnetization or because the magnetization intensities became too weak during demagnetization to be measured with a high sensitivity ScT cryogenic magnetometer (noise level less than

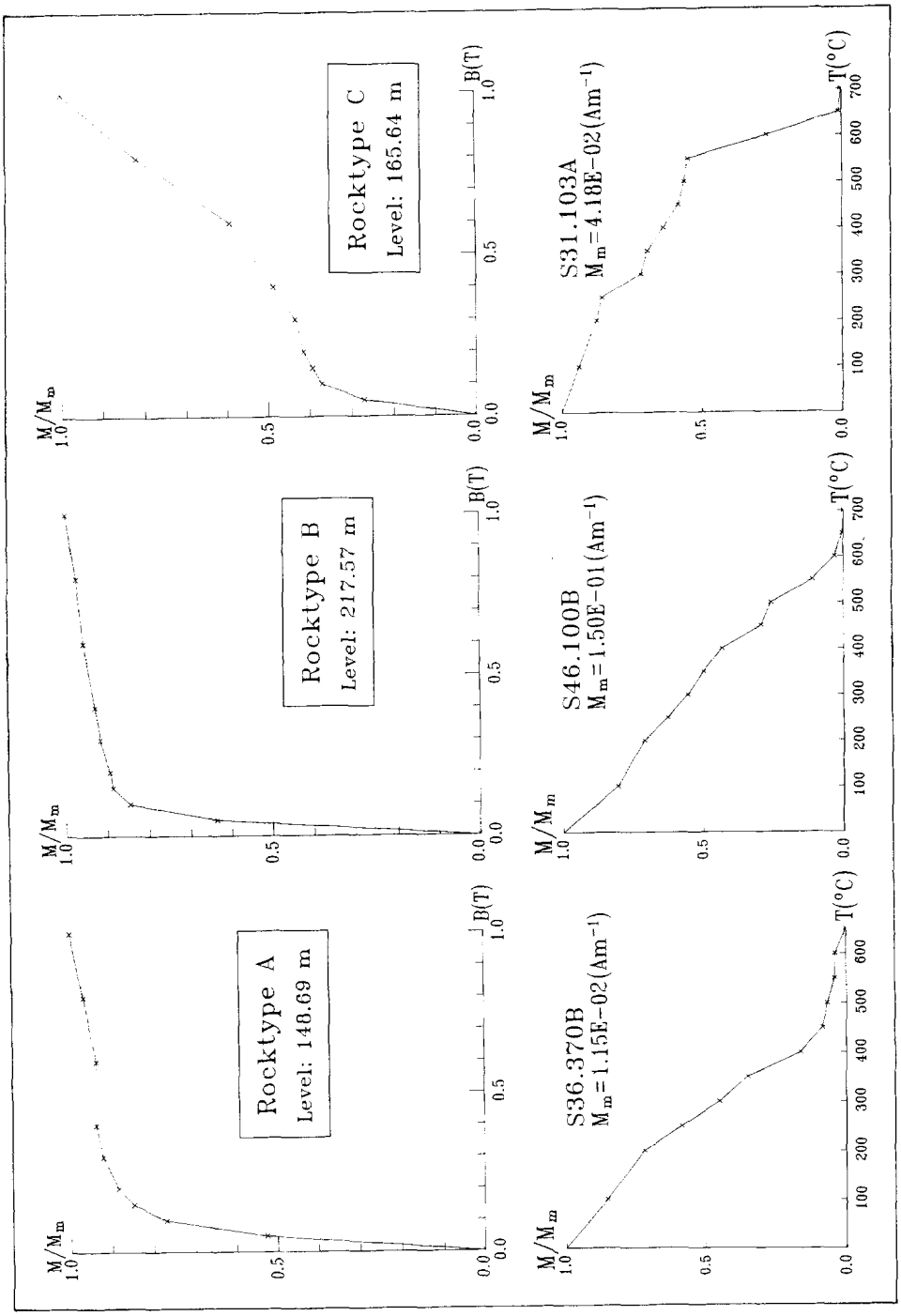



Fig. 4. IRM acquisition and thermal demagnetization of IRM of different limestones (rock types A, B, C).

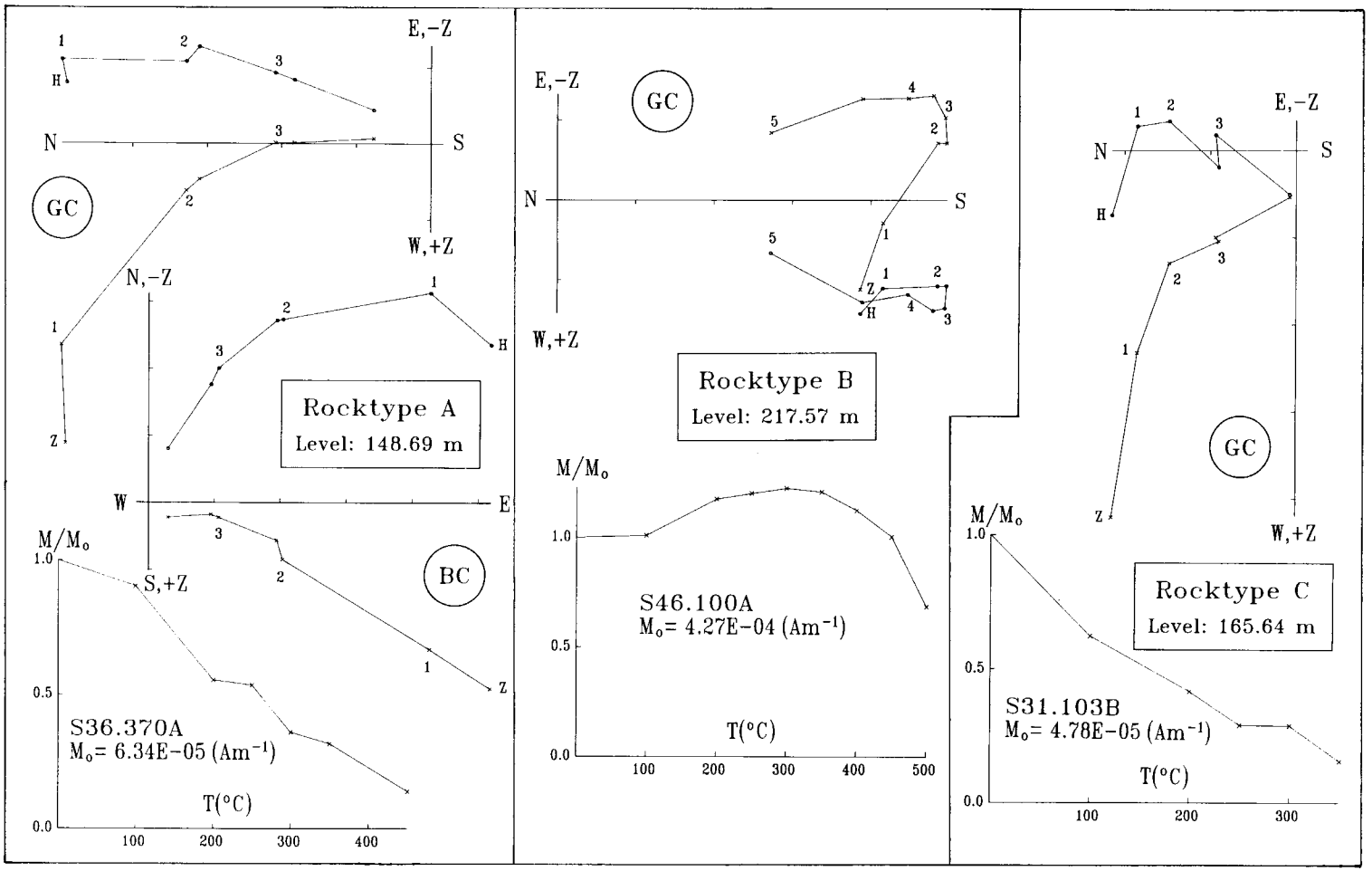



Fig. 5. NRM vectors (plotted on orthogonal vector diagrams) and intensities of the limestone rock types A, B, C during thermal treatment. *GC* denotes orthogonal vector plots in a geographical coordinate system, *BC* after bedding correction. Characteristic remanence (ChRM) components were determined by visual inspection and subsequent linear regression analysis of vector path segments trending towards or near the projection origin at elevated demagnetization temperatures.

0.005 mA $m^{-1}$). This led to unfortunately large sampling gaps in the Upper Permian, and also around the Griesbachian/Dienerian stage boundary (Fig. 2).

Three limestone rock types carrying a ChRM have been classified according to their magnetic characteristics (Figs. 4, 5). Type A builds up an IRM dominated by low coercivity: the IRM decays upon heating at temperatures between 400 °C and 450 °C, but a small part of the remanence survives and is only unblocked above 600 °C. Titaniferous magnetite (and/or maghemite) may carry the main part of magnetization unblocked below 450 °C and the remaining small contribution comes from haematite. Three components constitute the NRM vector. Up to 100 °C a component with variable direction is removed which probably originated during storage in the laboratory. The predominant direction, which parallels the present geomagnetic field at Guangyuan (Fig. 5, rock type A: projection in the geographic coordinate system, *GC*), is found between 100 °C and 300 °C. However, it is often possible to trace a very shallow, NNE aligned ChRM above 300 °C until the magnetization signal cannot be detected any more above 450 °C (Fig. 5, rock type A: projection after bedding correction, *BC*). Rock type A is observed mainly in the diamagnetic parts of the section (Fig. 2) where very pure or siliceous limestones have been deposited.

The coercivity spectrum of rock type B can be derived from the IRM acquisition curve and is very similar to the previous one. Again low coercivity minerals contribute most of the IRM. During thermal treatment this IRM is destroyed in two major steps, first up to 450 °C and then to 550–600 °C. This behaviour is explained by the presence of a titaniferous phase and a rather pure magnetite phase in addition. A small, high coercivity component carried by haematite with maximum unblocking temperatures higher than 600 °C, is superimposed. The NRM is composed of a present day vector, removed by 300 °C, and a ChRM vector which can be recognized in the sample of Fig. 5 (rock type B) as a reversed magnetization, inclined upwards at a shallow angle to the south. Above 550 °C the signal becomes erratic, probably due to thermochemical alteration of clay minerals present in this rock type. Rock type B is found mainly in marls, muddy limestones and mudstones around the P/Tr boundary and in the higher Feixianguan (Fig. 2). The susceptibilities are clearly positive and the remanence intensities (IRM, NRM) are an order of magnitude higher than in rock type A. The enlarged detrital input in these sediments obviously increases the amount of ferromagnetic constituents.

Rock type C (Figs. 4, 5) may be found at any stratigraphic level because it represents strongly weathered limestones. The example shown is an altered rock type A, the IRM of which is largely controlled by haematite of high coercivity and high unblocking temperatures. The NRM signal, however, disappears at 350 °C after removal of a characteristic component without any visible contribution from the haematite phase which in all three rock types is thought to be a product of recent weathering. The low NRM intensities of rock types A and C frequently prevented accurate determination of the directions of ChRM. This results in the large directional scatter observed in Figs. 2 and 3c, d.

## 3. Discussion

The inclination data in Fig. 2 contain appreciable uncertainty because of the difficulties in establishing the ChRM vector in many samples. No significant trend in inclination is present. The inclination (12.8°, Fig. 3d) of the vector mean direction for the Shangsi section gives a palaeolatitude at 6.5 °N. This agrees well with previous palaeomagnetic investigations [21–24] which have assigned a near-equatorial palaeogeographic position to the Yangtze block in the Upper Permian to Lower Triassic. It is also in close agreement with the observations of Opdyke et al. [3] and leaves a rather large latitudinal space between South China and Siberia which has been closed in post-Triassic times. However, the large intra-Permian apparent polar shift and the rapid northward movement of the South China block which was derived in Changxingian limestones by Lin et al. [23], is not supported.

Extremely weak and often unstable magnetization resulting in a rather incomplete sample profile were observed in the Upper Permian Dalong formation at Shangsi. Nevertheless the remaining samples considered to carry a ChRM characterize

the late Upper Permian as a time of predominantly normal polarity. The Kiaman interval of continued reversed polarity ended then and was superseded by the Illawarra interval of mixed polarity. The beginning of the Illawarra interval has been placed in the Upper Permian at the base of the Tatarian in Russia [15] which may correspond to the Wuchiapingian and Changxingian in China (A. Baud, in preparation), and at the end of the Upper Guadalupian in North America [14]. It was identified at the P/Tr boundary in South America [16]. Our results show that the Illawarra interval began earlier than the base of the sampled section (Wuchiapingian, Uppermost Permian), which has positive polarity and no longer belongs to the Kiaman interval.

Worldwide correlation of the polarity zones and assignment of a marker polarity event to the P/Tr boundary is not yet possible because of the poor age control in the mainly continental red sediments of the previous studies. In addition, the palaeontologically well dated polarity sequence at Shangsi is very much restricted in time covering only the Uppermost Permian to Lower Triassic. A complete polarity record of the Illawarra interval, however, may still be obtained in South China because continuous marine sedimentation started here already in the Carboniferous [1].

The sedimentation rate in the Shangsi section can only be crudely estimated because of (1) the problems of stratigraphic correlation between the systems used in different lands, and (2) inaccurate knowledge of the numerical ages of stratigraphic stages. The Lower Triassic sampled interval is about 80 m thick, the lower 52 m according to the stratigraphic description by Li et al. [17] being allocated to the Griesbachian, which lasted 1.25 m.y [25,26]. This gives an estimated sedimentation rate of about 42 m/m.y., which may be considered considerably in error and variable across the section, but no better definition can be expected soon.

The Shangsi limestone section contains 5 reversals of polarity, which make up a pattern of six distinct polarity zones (Fig. 2). Each is clearly defined by several samples; a single reversed sample is also found at about 178 m. The interpreted reversal rate of 1–2 reversals per m.y. is about half the average rate in the Late Cenozoic [27,28]. The shortest magnetozones are about 4.5 m thick, equivalent to a polarity interval duration of about 110,000 years.

According to Xu et al. [9] the P/Tr boundary in the Shangsi section is characterized by an iridium anomaly. An intriguing aspect of the magnetostratigraphy is the relation between the P/Tr boundary and the nearest polarity reversal. The last sample 15 cm below the boundary has negative polarity. Just above the boundary the rocks are marly, friable and weathered and could not be sampled satisfactorily. The next usable sample comes from 1.4 m above the P/Tr boundary, and it has positive polarity. Calculated with the mean sedimentation rate during the Griesbachian, the sampling gap represents about 35,000 years. However, the sedimentation rate is probably lower in the marly beds, and the time represented is correspondingly longer. The shortest magnetic polarity intervals known from analysis of Mesozoic and Cenozoic oceanic magnetic anomalies are thought to last for about 30,000 years. It is unlikely that the sampling gap conceals an additional pair of short magnetozones, but this is not impossible.

A striking feature of the magnetostratigraphy is the close correspondence between the P/Tr boundary and the following transition from negative to positive magnetic polarity. This resembles the situation for the Cretaceous/Tertiary boundary at Gubbio, Italy, and at DSDP site 524; the next reversal occurs some 80,000 years after the boundary in these sections [29]. The sampling gap in the Shangsi section makes a precise correlation impossible. Coincidence between the events cannot be ruled out, nor can it be established. The difference in time between the Permian/Triassic boundary and the next following magnetic transition is not greater than for the Cretaceous/Tertiary boundary.

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